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Code and data release · GitHub

github.com/brave1094/refine-first-trust-later

Refine First, Trust Later

**A guide to the repository — data, models,
pipeline, and results**

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Overview

Task-relative noise — definition, framework, five illusions

Definition

- Noise — a session without the task's target signal
 - e.g. OS background in an app class
- Task-relative
 - noise for one task, signal for another
- Measured per session
 - one column per rule

Framework

- 3 levels · 6 categories · 50 rules
- Mandatory — data quality
 - Aa collection defects (7)
 - Bb label errors (29)
- Optional — environment
 - Cc network control (5) · Cd background (4)
 - De third-party CDN/SSO (4) · Df browser (1)

Five illusions of noise

- Suppressed target signal
- Damaged learning dynamics
- Collapsed class boundaries
- Undermined evaluation objectivity
- Feature importance shifted to artifacts

⇒ **Refine first, trust later** — the repository lets anyone re-derive every number of the eight public datasets

Repository

Layout — assets, data, and one folder per stage of the pipeline

Folder	Contents	Step
00_assets/	datasets/ (captures + _tools/) · models/ (upstream code, weights)	1, 2
01_dataset/	session lists, session keys, model inputs, field-name maps	5, 6
02_preprocess/	sessions, statistics, 50-rule engine, lists, model inputs	3–6
03_model/	training, result tables; own_models/ (1,205 features, 2D-CNN)	T
04_analysis/	analyses of the five illusions, seed statistics	7
99_documents/	DATASETS · MODELS · PIPELINE · PAPER_MAP · guide · S1 · results/	—
repo_paths.py	the layout in one file · run_pipeline.sh: steps 3–7	—

Not included

- Captures
from their publishers
99_documents/DATASETS.md
- Private #1 and Private #2
no data, list, or result
- Model weights (~780 GB)
regenerated by training
- Per-session rule columns of cic18, iot23
regenerated by step 4

⇒ **One layout file** — every script takes its folders from repo_paths.py; released numbers sit in 99_documents/results/

Data — Step 1

Eight public datasets — official downloads and how each is arranged

ID	Dataset	Paper	Classes	Sessions	Arrangement (00_assets/datasets/_tools/)
vpn16	ISCX VPN-nonVPN 2016	Draper-Gil 2016	15	305,301	arrange_pcaps.py (keyword rules)
tor16	ISCX Tor-nonTor 2016	Lashkari 2017	15	62,593	arrange_pcaps.py (keyword rules)
tls1.3	CSTNET-TLS 1.3	Lin 2022	120	46,382	arrange_pcaps.py (domain folders)
cispec	CipherSpectrum	Wickramasinghe 2025	41	123,000	cispec/relabel_cispec.py
ustc16	USTC-TFC2016	Wang 2017	20	595,783	arrange_pcaps.py (Benign / Malware)
cic17	CIC-IDS2017	Sharafaldin 2018	15	1,571,244	arrange_pcaps.py + official flow labels
cic18	CSE-CIC-IDS2018	Sharafaldin 2018	14	51,234,354	cic18/apply_rename_map.py (4,006 names)
iot23	CIC IoT Dataset 2023	Neto 2023	30	1,155,285,412	arrange_pcaps.py (folder names)

※ Classes (task 3) and sessions before refinement, as in Table 1 of the article. cispec: the released archive holds 41 domains (the paper and page state 40).

⇒ **Captures are not redistributed** — download from the publishers; links, files, and terms in 99_documents/DATASETS.md

Data — Step 1

Check and align — released lists work with any file names



Content key of a session

- transport | sorted endpoints | first-packet time (μ s)
independent of file names and of the tshark version
- Keys of all listed sessions released
01_dataset/<ds>/session_keys.csv — 7 datasets, iot23 to follow
- Self-check
our data aligned to itself: every row of every list identical

tor16 — identical copies in the release

- Same session up to four times
7–12 % of the test lists have a copy in training
- Illusion 1 without them
 $\Delta +2.15 \rightarrow +1.96$ pp, 95 % CI [1.38, 2.53]
all seven models still positive

⇒ **Alignment by content** — reproduces the sessions of the article even when file names differ

Models — Step 2

Seven models — upstream code pinned, not redistributed

Model	Paper	Code	License	Our change
XGBoost	Chen 2016	<code>pip xgboost</code>	Apache-2.0	1,205 statistical features (ours)
Random Forest	Breiman 2001	<code>pip scikit-learn</code>	BSD-3	same 1,205 features
2D-CNN	Wang 2017	03_model/own_models (ours)	MIT	own implementation
ET-BERT	Lin 2022	linwhitehat/ET-BERT @ d594706	MIT	none
YaTC	Zhao 2023	NSSL-SJTU/YaTC @ a220b75	no license file	4-line forward() shim ($\text{timm} \geq 0.9$)
NetMamba	Wang 2024	wangtz19/NetMamba @ bef641e	no license file	none
TrafficFormer	Zhou 2025	IDP-code/TrafficFormer @ 6d0ba64	MIT	none

※ Weights: links of the upstream READMEs, listed in 99_documents/MODELS.md. Repositories without a license file are fetched from their authors, never copied here.

⇒ **bash 00_assets/models/setup_models.sh** — clones each pinned commit (169 files identical to our copies) and fetches the weights

Pipeline — Steps 3 to 7

Steps in order — `bash run_pipeline.sh <step> [datasets]`

3	Sessions and statistics	01_session_split.py · 03_session_stat.py · official labels (cic17, cic18)	03_session_stat/
4	Rule marking	04_noise_labeling.py — one column per rule, every session	04_session_noisy_labeled/
5	Session lists	released lists (align) or 05_make_filelist.py · 07_* (1,000 per class, 8:2)	01_dataset/<ds>/00_filelist*
6	Model inputs	06_make_dataset.py — 7 models × 3 variants, IP/port masking	01_dataset/<ds>/<variant>/
T	Training	01_train.py — Exp1–Exp4 × 3 variants × 5 seeds × 7 models	99_documents/results/
7	Analyses	04_analysis/ — a3–a7 · 01–05 · seed_stats (PAPER_MAP.md)	99_documents/results/

⇒ **Settings of the article built in** — batch sizes, epochs, Optuna trials, and seeds 42 · 1 · 7 · 2024 · 31337

Pipeline — Steps 3 to 7

Experimental design — four conditions and three list variants

	Training set	Test set	Used for
Exp1	refined	refined	Illusions 1, 2, 3, 5
Exp2	refined	noisy	Illusion 4
Exp3	noisy	refined	Illusions 1, 2, 3, 5
Exp4	noisy	noisy	usual practice; Illusion 4

※ In the code, denoised = refined (the article's term).

※ Illusion 1 compares Exp1 with Exp3 on the same refined test set.

Three list variants

- full
refined and noisy lists drawn independently
up to 1,000 sessions per class, hash split 8:2
- sizectrl — main numbers
noisy training list cut to the refined size
- sizectrl+strat
up to 5 pp noise injected per class, paired

⇒ **One comparison per illusion** — only the training set changes between Exp1 and Exp3, only the test set between Exp2 and Exp4

Results and Reproduction

Released numbers — Illusion 1 on the eight public datasets (sizectrl, 5 seeds × 7 models)

Dataset	$\Delta = \text{Exp1} - \text{Exp3}$ (pp)	95 % CI	Models +/-	Verdict
vpn16	+17.73	[+17.00, +18.45]	7 / 0	significant (+)
tor16	+2.15	[+1.68, +2.61]	7 / 0	significant (+)
tls1.3	-0.19	[-0.41, +0.03]	3 / 4	within error
cispec	+0.03	[-0.13, +0.18]	3 / 0	within error
ustc16	-0.04	[-0.18, +0.09]	2 / 5	within error
cic17	-0.23	[-0.76, +0.31]	2 / 4	within error
cic18	+1.10	[-1.23, +3.44]	2 / 4	within error
iot23	+6.60	[+5.48, +7.72]	7 / 0	significant (+)

Reading

- Significant where sampled noise is high
vpn16 73.6 % · iot23 27.5 % · tor16 26.6 %
- Within error on low-noise datasets
tls1.3 · cispec · ustc16 · cic17 · cic18
- Source file
99_documents/results/analysis/
08_seed_stats/sizectrl/seed_stats_ds.csv

⇒ **Every public number is a file** — 99_documents/PAPER_MAP.md maps each table and figure to its script and CSV

Results and Reproduction

Quick start — one dataset from download to analysis

```
git clone https://github.com/brave1094/refine-first-trust-later.git
cd refine-first-trust-later && pip install -r requirements.txt
bash 00_assets/models/setup_models.sh — step 2 — upstream code, weights
python 00_assets/datasets/_tools/arrange_pcaps.py --dataset vpn16 --src <dl> — step 1
bash run_pipeline.sh 3 vpn16 — sessions, statistics, check
bash run_pipeline.sh align vpn16 — released lists
bash run_pipeline.sh 4 vpn16 — 50-rule marking
bash run_pipeline.sh 6 vpn16 — model inputs
GPU=0 bash run_pipeline.sh train vpn16 — Exp1–Exp4
bash run_pipeline.sh 7 — analyses
```

Where to read

- 99_documents/DATASETS.md
downloads, arrangement, checks
- 99_documents/MODELS.md
commits, weights, settings
- 99_documents/PIPELINE.md
every step and its output
- 99_documents/PAPER_MAP.md
article → script → CSV

⇒ **Requirements** — Python 3.10+, Wireshark (tshark, editcap), and the environment of each upstream model



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Thank you

Refine first, trust later — reproduce, extend, and question our numbers.

github.com/brave1094/refine-first-trust-later

Code MIT · data CC BY 4.0 · cite via CITATION.cff

